Proposal Defense Kit

Group 4 - Predicting At-Risk Students - IC-ML Machine Learning Capstone

Speaker script and panel Q&A prep. Target: ~10 minutes. Six members can split the slides; aim for roughly 30 seconds per slide and do not read the bullets - say the idea in your own words.

Part 1- Speaker Script (slide by slide)

Slide 1- Title (~20s) Speaker: _____

Good morning/afternoon, panel. We are Group 4. Our capstone proposal is "Predicting At-Risk Students from Social and Behavioral Factors: A Machine Learning Approach to Early Academic Intervention." I am [name], together with [members].

Slide 2- Why Academic Success Matters (~30s) Speaker: _____

We want to start with why this matters. A student's academic performance strongly shapes their future opportunities. But in every class, some students quietly fall behind, and whether they get help depends on someone noticing in time. So the question we kept coming back to is: how do we know who is struggling before it is too late?

Slide 3- The Problem: Risk Is Seen Too Late (~35s) Speaker: _____

Right now, the clearest signal that a student is in trouble is the final grade. But by the time it arrives, the term is over and it is too late to intervene. And a single teacher handling dozens of students cannot realistically track each one's risk. So the support that does happen tends to come only after the student has already failed.

Slide 4- The Cost of Late Detection (~30s) Speaker: _____

That late detection has real costs: a missed window for tutoring or counseling, grade retention - repeating the subject or the year - and a higher chance the student disengages or drops out entirely.

Slide 5- How Is It Handled Now? (~30s) Speaker: _____

Currently, support is mostly reactive and based on manual observation, which is limited by class size. There is no systematic early signal. Which leads to our core question: can machine learning help identify at-risk students early?

Slide 6- Related Work (~40s) Speaker: _____

This is not a brand-new idea. Prior studies - Cortez and Silva, Yagci, and several review papers - show that student outcomes are learnable from demographic, social, and behavioral data using classifiers. But we noticed a gap: many of them lean heavily on prior grades as inputs, and few actually measure how much those grades contribute versus predicting without them. That gap is what we build on.

Slide 7- Problem Statement (~25s) Speaker: _____

So, our problem statement: academic risk is usually identified only at the end of the term, from grades that come too late to act on. There is a need to identify at-risk students earlier, so that support can be given while it still matters.

Slide 8- Motivation (~25s) Speaker: _____

Our motivation is to build a data-driven early-warning tool - one that flags students at risk of failing so that teachers and guidance staff can prioritize who needs help, before the finals.

Slide 9- Dataset: Rationale and Overview (~35s) Speaker: _____

For data, we use the UCI Student Performance dataset, the mathematics subset: 395 students, 33 attributes, framed as a binary pass/fail problem. About one in three students fail, so the classes are imbalanced - that matters later. We chose it because it is public, reliable, rich in social and behavioral factors, and a well-known benchmark.

Slide 10- Dataset: Key Terminology (~30s) Speaker: _____

Quick on terminology: the 33 attributes fall into four groups - demographic, family background, academic history, and behavioral or social. Our target: a student passes if the final grade, G3, is 10 or above, and is "at risk" if it is below 10.

Slide 11- Understanding the Grades (~40s) Speaker: _____

A key part of our data is the three grades. The dataset is from Portuguese schools, which grade in three periods. G1 and G2 are like our prelim and midterm grades - they hint at the outcome, but the final grade, G3, is the real result. And they are not destiny: in our data, 24 students

recovered from a failing second period, while 8 who were passing still failed. So G3 is a genuine, separate outcome.

Slide 12- The Role of G1 and G2 (~45s) Speaker: _____

This raised a design question. G1 and G2 are very strong predictors - the second-period grade alone matches the final 92% of the time. So instead of just throwing them in or throwing them out, we test both. Scenario 1 includes them, reflecting what a teacher knows by mid-year. Scenario 2 excludes them, a harder and earlier prediction from stable factors only. Comparing the two tells us how much the interim grades actually add, and how early we can still predict.

Slide 13- General Objective (~25s) Speaker: _____

Our general objective: to develop and evaluate a machine learning classification solution that identifies students at risk of failing the final grade, using their academic, social, and behavioral attributes, to support timely intervention.

Slide 14- Specific Objectives (~40s) Speaker: _____

Specifically, five objectives: train and compare three classifiers; compare performance with and without the interim grades; apply feature selection and dimensionality reduction to find the most influential factors; evaluate rigorously, prioritizing recall and F1 on the at-risk group; and deliver a documented, reproducible Python prototype.

Slide 15- Proposed Methodology (~35s) Speaker: _____

Our methodology is a standard supervised-learning pipeline: data acquisition, preprocessing, model development, feature selection and PCA, evaluation, and a prototype demo. It is all implemented in Python with Spyder, using pandas, scikit-learn, and Matplotlib and Seaborn.

Slide 16- Proposed Models and Evaluation (~40s) Speaker: _____

For models we use the three classifiers from our course module: logistic regression, k-nearest neighbors, and the support vector machine. For evaluation, because the classes are imbalanced, we focus on recall - catching the students who actually fail - along with precision, F1, the confusion matrix, and ROC-AUC. Accuracy alone can be misleading here, so we do not rely on it.

Slide 17- Expected Output and Contributions (~30s) Speaker: _____

We expect to deliver: a documented Python pipeline; a comparison of the three classifiers; a measure of how much G1 and G2 improve prediction; a ranked list of the factors most tied to risk; and a validated early-warning prototype.

Slide 18- Projected Timeline (~25s) Speaker: _____

Our timeline runs over the term in six phases - from data acquisition and exploratory analysis, through preprocessing, model development, feature selection and tuning, evaluation, and finally the paper and the presentation.

Slide 19- Thank You (~15s) Speaker: _____

That is our proposal. Thank you - we are happy to take your questions.

Running total: about 9 minutes, leaving buffer for pacing and transitions.

Delivery tips

- Do not read the slides. The bullets are for the panel; you say the idea.

- Hand off cleanly between speakers ("...and [name] will take the dataset").

- Slow down on Slides 11-12 (the G1/G2 story) - that is your strongest point and the most likely to be questioned.

- Keep an eye on time; if you are running long, trim Slides 4 and 5.

- End on time and invite questions confidently.

Part 2- Panel Q&A Prep

Anticipated questions with ready answers. Stay calm, acknowledge good questions, and it is fine to say "we will explore that further in the finals."

Q1. Isn't predicting the final grade from the earlier grades (G2) trivial, or data leakage?

A: Fair point - and it is exactly why we test both scenarios instead of just including them. Yes, G2 correlates about 0.90 with G3, but it is not trivial: only 19% of students have G3 exactly equal to the average of G1 and G2, and 32 students flip outcome - 24 recover, 8 collapse. Scenario 1 is a realistic progress-monitoring use, and Scenario 2 removes the grades entirely to show how early we can predict. We measure the contribution rather than hide behind it.

Q2. Why only three classifiers? Why not Random Forest or neural networks?

A: We deliberately scoped to the three classifiers formally taught in our course module - k-nearest neighbors, logistic regression, and SVM - so the work stays within what we can rigorously justify and implement. They cover three distinct model families: distance-based, linear/probabilistic, and margin-based. Ensembles like Random Forest are a natural extension if time allows in the finals.

Q3. Why classification instead of regression (predicting the exact grade)?

A: Our goal is early identification of at-risk students, which is inherently a pass/fail decision - that maps to classification. Predicting the exact 0-to-20 grade from these factors is both harder and less actionable for intervention. Classification also gives intuitive, decision-focused metrics like recall and the confusion matrix.

Q4. Are you developing or building the machine learning algorithms?

A: No. We use established algorithms from scikit-learn. What we build are the trained models, the comparison between them, and the applied pipeline. Our contribution is the application and the with/without-grades analysis, not a new algorithm.

Q5. Why prioritize recall over accuracy?

A: Because the classes are imbalanced - about one in three fail. A model that predicts "everyone passes" scores about 67% accuracy but catches zero at-risk students, which is useless. Recall measures how many actual at-risk students we catch, which is the whole point, and F1 balances that against false alarms.

Q6. Why this dataset? Is it relevant to our context?

A: It is a public, reliable, widely benchmarked dataset with rich social and behavioral attributes, which lets us focus on the methodology. The factors - study time, past failures, absences, family support - are general enough to transfer, and the method, not the specific data, is the deliverable.

Q7. How do you handle the class imbalance?

A: Two ways. We evaluate with imbalance-appropriate metrics - recall, F1, the confusion matrix, and ROC-AUC - instead of accuracy, and we use stratified splitting and class weighting so the models do not simply favor the majority class.

Q8. What are the limitations?

A: The dataset is small (395 records), which limits model complexity; the at-risk class is a minority; and without the interim grades the accuracy will be lower - but that gap is a result we want to measure, not a flaw.

Q9. What is your expected or target accuracy?

A: We are not committing to a fixed number in the proposal, because it depends on the scenario. With the interim grades we expect high performance; without them, lower but still useful. The comparison itself is the finding.

Q10. How is this different from the existing studies you cited?

A: Most include prior grades and report high accuracy without questioning it. We explicitly compare the same pipeline with and without those grades, so we quantify their contribution and test how early prediction is feasible.

Q11. Some students have a final grade of 0 - what are those?

A: About 38 students have a final grade of zero, likely dropouts or no-shows. They are extreme at-risk cases; we handle them in preprocessing and exploratory analysis and will note their effect on the results.

Q12. What exactly will you deliver for the finals?

A: A working Python and Spyder implementation of the full pipeline, the trained and compared models under both scenarios, evaluation results and visualizations, and the final paper.

If you get a question you do not know

Acknowledge it, give your best reasoning, and offer to address it in the finals: "That is a good point - our current plan is X, and we will look into that during implementation." Confidence and honesty beat guessing.